

Full operator image in finite dimension

`MathlibAnnex.Analysis.CStarAlgebra.Representation.surjective_of_irreducible_of_finiteDimensional`

theorem

Constructs a preimage of an arbitrary operator by separately lifting its self-adjoint real and imaginary parts.

Statement

Let A be a unital complex C^* -algebra and H a nonzero finite-dimensional complex Hilbert space. If $\pi : A \rightarrow \mathcal{B}(H)$ is an irreducible unital representation, then

$$\forall T \in \mathcal{B}(H) \exists c \in A : \pi(c) = T.$$

Assumptions

No singleton condition, faithfulness or separability of A is assumed. The finite-dimensionality and nontriviality hypotheses concern H .

Conclusion

The image of π is all bounded operators on H .

Proof route

Interpolate each self-adjoint operator on the whole finite-dimensional space, then use the real-imaginary decomposition.

▼ Proof steps

1. For a self-adjoint bounded operator S , apply Self-adjoint interpolation on a finite-dimensional subspace to the irreducible π , the finite-dimensional subspace $E = H$, and this S . The theorem supplies a self-adjoint $a \in A$ with $\pi(a)x = Sx$ for every $x \in E$ (and a norm bound which is not needed here). Since $E = H$, equality on all vectors gives $\pi(a) = S$. This is packaged in Preimages of self-adjoint operators on the whole finite-dimensional space.
2. For an arbitrary T , define

$$\operatorname{Re} T = (T + T^*)/2, \quad \operatorname{Im} T = (T - T^*)/(2i).$$

Both operators are self-adjoint, so Step 1 gives $a, b \in A$ with $\pi(a) = \operatorname{Re} T$ and $\pi(b) = \operatorname{Im} T$.

3. Use the same two preimages to set $c = a + ib$. Complex linearity gives

$$\pi(c) = \pi(a) + i\pi(b) = \operatorname{Re} T + i \operatorname{Im} T = T.$$

This constructs the required preimage of each T .

Main citations

- Exact theorem and proof
- Self-adjoint interpolation on a finite-dimensional subspace
- Preimages of self-adjoint operators on the whole finite-dimensional space

Lean source signature (exact)

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theorem surjective_of_irreducible_of_finiteDimensional [Nontrivial H]
  [FiniteDimensional ℂ H]
  (pi : Representation A H) (hirr : pi.IsIrreducible) :
  Function.Surjective pi
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In the source	Mathematical meaning
[Nontrivial H]	The representation space is nonzero.
[FiniteDimensional ℂ H]	The entire Hilbert space has a finite complex basis.
(pi : Representation A H)	The unital $\ast$ -representation of the ordered unital $C^\ast$ -algebra A .
(hirr : pi.IsIrreducible)	This representation is nonzero and has no proper nonzero closed reducing subspace.
Function.Surjective pi	For every bounded complex-linear operator $T : H \rightarrow H$, some algebra element c satisfies $\pi(c) = T$ as operators.

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

Declaration: MathlibAnnex.Analysis.CStarAlgebra.Representation.surjective_of_irreducible_of_finiteDimensional

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Accepted source guide SHA-256: edec39f02355afccce3e457cbbea41368524f4a5fb261ac23d30c7fbeatdc2b9

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73